

A Stiffness-Adaptive Visuo-Tactile Robotic Palpation System for Surface Texture Profiling and Deep Lesion Localization

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Abstract—Accurate intraoperative tissue characterization relies on the ability to resolve both surface micro-textures and deep-seated structures for comprehensive diagnosis. However, achieving surface profiling and deep-tissue inspection remains a critical challenge due to the difficulty of combining variable stiffness and high-fidelity perception within a compact device. Here, we propose a robotic palpation system consisting of a novel visuo-tactile sensing probe (diameter: 10 mm) integrated into a robotic platform for accurate tissue characterization. This probe exhibits adaptive strain-stiffening behavior (achieving a 470-fold stiffness variation), applicable for various surgical conditions. With our 3-D reconstruction and scanning framework, the palpation system achieves high-precision 3-D reconstruction (MAE: 28 μm in the z -axis, 3.1 μm in the xy -plane), profiling surface textures (feature size: 150 μm), and delineating the boundaries of complex deep-seated inclusions (sensitivity > 0.92). Finally, an *ex vivo* porcine liver model was prepared to demonstrate our system's capability to analyze and decouple microscale superficial irregularities from macroscale lobulated deep-seated lesions under

dynamic disturbances. These results confirm the system's potential as a precise, quantitative platform for robotic surgical palpation in complex pathological scenarios.

Index Terms—Medical robots, robotic palpation, tactile sensing, variable stiffness, visuo-tactile sensors.

I. INTRODUCTION

MANUAL palpation is widely used as the primary examination method for tissue characterization in traditional clinical practice. Surgeons typically perform a continuous palpation process: global surface sweeping to locate suspicious anomalies, followed by localized depth indentations using their fingers to characterize the stiffness and boundaries of underlying lesions. Over the past two decades, robot-assisted interventions have been deployed in various clinical procedures in general surgery [1], [2], [3], [4]. Despite the rapid development of minimally invasive surgery (MIS), most robotic platforms rely on noncontact visual endoscopes [5]. Without quantitative haptic feedback, these systems are not only susceptible to interferences such as blood and dynamic occlusion but also fail to perceive subsurface mechanical properties, such as tumor stiffness and 3-D boundaries. From an engineering perspective, a robotic palpation system must provide an objective, quantitative mechatronic measurement process to resolve mechanically coupled, multiscale pathological challenges. The palpation system should be able to resolve superficial irregularities, such as the fibrotic micronodules in early-stage cirrhosis [6] and the miliary peritoneal implants with ovarian metastasis [7]. It also requires penetration of tissues to localize deep-seated tumors, e.g., endophytic tumors that are embedded within tissues [8]. Hence, advancing MIS requires a high-precision quantitative robotic platform capable of decoupling superficial irregularities from deep subsurface features for precise surgical navigation.

Integrating tactile sensing into surgical end-effectors is the primary strategy to endow the haptic feedback during surgical operations. Commercial force/torque sensors are the most prevalent solution for robotic palpation [9], [10]. These strain-gauge-based sensors are widely employed to measure local tissue stiffness via contact force and indentation depth analysis. Although these sensors facilitate the generation of global stiffness maps through continuous robotic scanning [9], [11],

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their rigid and bulky structures are incompatible for operation in confined surgical spaces. To overcome dimensional constraints and enhance sensory feedback, researchers have employed fiber Bragg grating sensors [12], [13], planar capacitive sensor patterns [14], [15], and vision-based tactile sensors [16] to achieve compact integration and multimodal perception. These sensors have demonstrated modulus estimation and localization during robotic palpation; however, they are limited to discrete single-point sensing feedback, suffering from low scanning accuracy.

To address the limitations of rigid probes restricted to single-point contact, efforts have been devoted to flexible sensing arrays. Recent advances in tactile sensing have focused on developing flexible electronic and magnetic skins, which leverage high-density and force-decoupled sensing, enabling comprehensive tactile perception [17], [18]. Electric-based sensing principles have been employed for various medical applications, ranging from smart surgical grippers [19] and robotic guidewires [20]. Moreover, biomimetic capacitive arrays utilize pyramid microstructures, which are designed to enhance sensitivity and mechanical stability [21]. Distributed sensing networks with resistive microchannels, resembling 3-D innervation structures, were developed to decouple complex external stimuli [22]. However, the current sensing array relies on complex wiring, which imposes constraints within the limited space of surgical instruments.

Vision-based tactile sensors offer a viable approach to surface analysis with a microscale spatial resolution [23], [24]. The integration of the imaging unit eliminates the complex wiring of a discrete sensing strategy, presenting great potential for robotic palpation. Conventional visuo-tactile sensors rely on photometric stereo to reconstruct 3-D topography via multidirectional illumination. These systems generally utilize a solid elastomer suspended over an air cavity as the contact interface. To minimize sensor dimensions, optical fiber bundles were employed for transmitting illumination and imaging with a diameter of 15 mm, separating the sensing interface from the supporting circuitry [25]. However, optical fiber bundles limit the spatial resolution of the system. In addition, an ultracompact vision-based sensor with a diameter of 8 mm was realized by employing a mechanoresponsive elastomer for depth analysis [26]. Still, the sensor presents insufficient spatial resolution due to the nonlinear color variation of the stimulated material response.

In clinical scenarios, existing tactile sensors in robotic palpation focus on the localization of deep-seated tumors via bulk stiffness estimation, suffering from coarse spatial resolution and poor conformability for surface tissue inspection. Conventional vision often fails to resolve textural pathologies due to poor contrast and specular reflections. To overcome this challenge, the robotic palpation system should allow for characterizing rigid micronodules on the soft tissue surface and rigid tumors embedded in tissue simultaneously. However, realizing this capability presents an inherent conflict: the sensor interface must be highly compliant for texture profiling, yet sufficiently rigid to deform and analyze deep tissues.

In response to these conflicting stiffness requirements, various variable-stiffness mechanisms [27], such as granular jamming

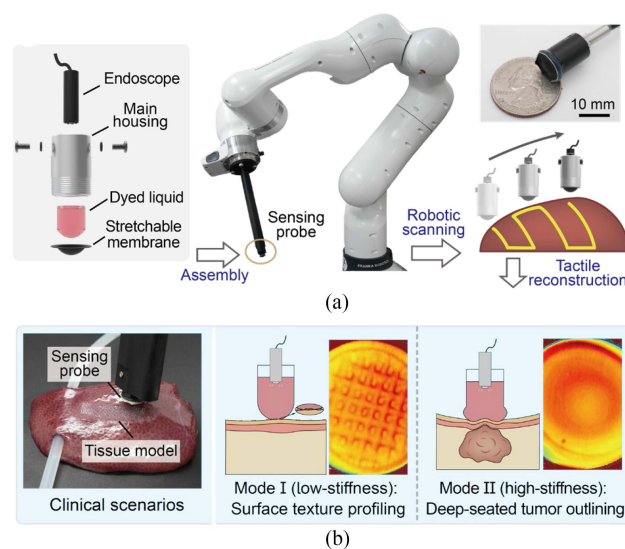


Fig. 1. Overview of the robotic palpation system. (a) Composition of the robotic platform, sensing probe, and scanning process. (b) Two palpation modes of surface profiling and subsurface tracking.

[28], layer jamming [29], and fluidic modulation techniques [30], have been explored in flexible devices. However, implementing these mechanisms requires bulky actuation systems or complex active material structures, making them incompatible within miniature end-effectors. To circumvent these limits, passive variable-stiffness approaches, such as shear stiffening gels [31] and adaptive structures [32], eliminate external tethers, yet they still require specific triggering conditions or specialized structural designs for stiffness tuning. Therefore, an urgent demand for a tactile sensing system with tunable stiffness capable of characterizing superficial microtextures and identifying deep-seated tumors to advance robotic palpation in MIS.

To achieve variable stiffness and high-fidelity perception during tissue inspection, we propose a novel visuo-tactile robotic palpation system, comprising a stiffness-adaptive sensing probe and a robotic platform for surface texture profiling and deep lesion localization (see Fig. 1). The main contributions of this work are as follows.

- 1) *A compact, integrated high-precision sensing unit:* We developed a miniature visuo-tactile unit with a diameter of 10 mm and a length of 25 mm. By employing a spectral filtering model for 3-D reconstruction, this design integrates an endoscope for imaging and illumination. The integrated system achieves high-precision 3-D reconstruction with an MAE of $28 \mu\text{m}$ in the z -axis and $3.1 \mu\text{m}$ in the xy -plane.
- 2) *A passive stiffness-adaptive mechanism via fluid-structure interaction:* We proposed a compact sensing structure, composed of a stretchable membrane enclosing a fluidic medium. Through fluid-structure coupling, the sensing probe achieves a 470-fold passive stiffness variation under external loading, enabling texture profiling and deep tumor analysis by two interaction modes.
- 3) *High-fidelity robotic palpation of various pathologies:* We integrated the sensing probe into a robotic platform

to establish a palpation scanning framework for accurate inspection. This system enables high-fidelity characterization of fine surface textures and boundaries of deep-seated inclusions and presents the capacity to decouple surface and subsurface anomalies, validating its potential for versatile pathological assessments.

II. MIS-COMPATIBLE ROBOTIC PALPATION SYSTEM DESIGN

This section presents the comprehensive design of the robotic palpation system to address the dimensional and functional constraints of MIS. First, the structural design, fabrication process, and functions of the sensing probe are described. Then the system integration is presented, combining the probe with a robotic platform to perform various palpation tasks [see Fig. 1(a)].

A. Fluid-Structure-Coupled Visuo-Tactile Probe

To meet the requirements of modern MIS, the design of tactile sensors should consider the challenges in miniature dimensions, stiffness tunability, and high-density sensing capacity. As compared in Table S1 of the Supplementary Material, existing miniature visuo-tactile sensors generally suffer from tradeoffs among these critical requirements. To overcome this limitation, our probe features an integrated optomechanical structure without external functional devices. We analyzed the design details for the liquid-structure coupling interface, the compact optical system, and the robotic scanning platform.

1) *Stretchable Bilayer Membrane*: A stretchable membrane serves as the primary sensing interface at the distal tip of the sensing probe. The membrane is fabricated with a silicone bilayer structure. The proximal surface is designed with a microscale roughness to induce a Lambertian diffusive effect, and the distal interface possesses a smooth surface to optimize optical reflection and enhance the reflected light intensity. This outer layer also functions as a protective barrier, shielding against ambient light interference and enhancing the durability during contact with biological tissues. Furthermore, the membrane thickness should be thin enough to resolve surface textures and possess robustness to support the high contact forces for tumor localization.

The bilayer membrane was fabricated via a two-step molding process. First, a white silicone precursor (Ecoflex 00-30 mixed with white pigment at a 1:1:0.35 mass ratio) was injected between a release-agent-treated 1000-mesh matte glass substrate and a top glass plate, spaced by 0.2 mm spacers. Upon curing, a black protective layer (Ecoflex 00-30 with black pigment, identical ratio) was cast directly onto the white layer following the same protocol to ensure uniform thickness.

2) *Dyed Liquid*: The dyed liquid works as the optical sensing medium and stiffness-tuning agent in this tactile sensing probe. Moreover, the stretchable membrane and dyed liquid constitute an active sensing medium rather than a passive compliant buffer. To transform standard RGB images into quantitative 3-D RGB-D data, we employed our previous sensing model to build a linear relationship between output color and length [33]. The aqueous red dye (Zhanyi Company) was selected to align with the color response of the camera for high-precision measurement. The

spatial resolution along the z -axis is determined by the dye concentration. The hyperelasticity of the membrane induces a strain-stiffening effect, which serves for the system's stiffness-tuning capability. The dyed liquid was prepared by mixing the deionized water with the aqueous dye at volume ratios from 1600:1 to 800:1 to generate a stable solution.

3) *Illumination and Imaging*: Unlike conventional visuo-tactile sensors that require a complex array of RGB LEDs, our system utilized a standard commercial endoscope for integrated illumination and imaging. In the illustration system, the white LED provides a continuous emission spectrum. Given the negligible LED-camera baseline in the compact endoscope, we simplify the optical model using a coaxial approximation. The illumination and detection paths are treated as collinear. Furthermore, assuming the sensing membrane exhibits ideal diffuse reflection, a robust one-to-one spatial correspondence is established between the CMOS of the camera and the membrane surface. The endoscope features a 3.9 mm outer diameter with a resolution of 1280×720 . The field of view of the camera module is 88° , and the depth of field ranges from 5 to 30 mm, ensuring that the entire surface is in focus.

B. Robotic Platform Assembly

The robotic palpation system utilizes a high-precision motion platform for precise manipulation tasks. This platform can eliminate mechanical vibrations during surface contact, providing a stable kinematic structure for data acquisition. The tactile sensing probe is attached to the system via a customized 3D-printed mount. To complement the local tactile sensing with global visual feedback, a monocular camera is integrated onto the side of the sensing probe to facilitate autonomous navigation toward the region of interest (ROI). This setup establishes an integrated visuo-tactile perception system, in which the camera guides the probe to the target region, enabling the subsequent high-resolution tactile exploration.

III. CONTACT MECHANICS AND STIFFNESS MODELING OF THE SENSING PROBE

This section develops an analytical model to describe the fluid-structure interaction between the internal incompressible fluid and the stretchable membrane based on continuum mechanics. By combining the geometric nonlinearity of the contact area and the fluid's volume conservation, we derive the relationship between the applied load and the variable stiffness. This theoretical model is essential for understanding the probe's sensing capabilities for various conditions and guiding its control strategies in palpation tasks.

A. Load-Adaptive Dual-Mode Perception

Mode I is in the bending stiffness-dominated regime for light-touch profiling. At low contact loads, the mechanical response of the membrane is dominated by local bending stiffness. The noncontact region remains stable without bulging, enabling high-fidelity reconstruction of surface microtextures without compressing the tissue.

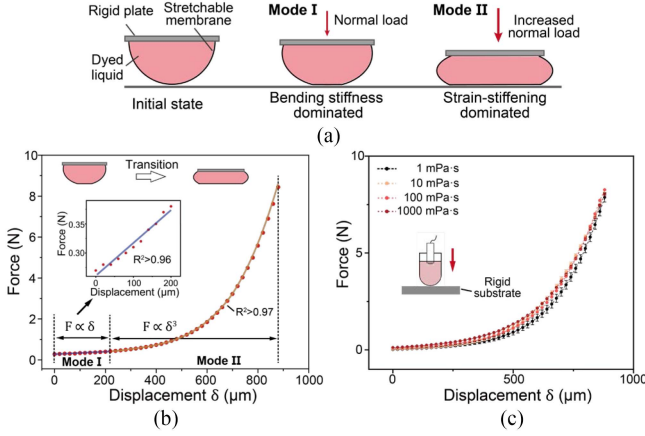


Fig. 2. Stiffness response of the sensing probe. (a) Schematic of the probe in two stiffness regimes. (b) Experimental characterization of the adaptive strain-stiffening behavior. The inset shows the linear response of Mode I. (c) Effect of internal liquid viscosity on stiffness.

Mode II is in the inflation-dominated regime for deep-indentation palpation. At higher loads, the incompressible fluid transfers the displaced volume to the noncontact region, causing elastic expansion and generating a bulge. This large deformation induces a strain-stiffening effect within the membrane, actively tuning the effective stiffness of the probe. This mechanism allows the sensing interface to indent into the soft tissue to reveal the deep-seated structure [see Fig. 2(a)].

B. Contact Mechanics of the Sensing Probe

To validate the sensing performance and variable stiffness of the probe, we build a contact mechanics model to analyze the force response in combination with membrane theory, fluid incompressibility, and Pascal's principle.

Based on the deformation condition, and considering it as a series system, the total displacement δ consists of δ_p from the probe's compression and δ_s from measured sample

$$\delta = \delta_p + \delta_s. \quad (1)$$

Assuming a flexible spherical cap contact with a rigid plate ($a \ll R$)

$$\delta_p = R - \sqrt{R^2 - a^2} \approx \frac{a^2}{2R} \quad (2)$$

where a is the contact radius, and R is the initial radius of curvature of the probe.

Treating the target tissue as an elastic body, the central displacement is given by

$$\delta_s = \frac{2a(1 - \nu_s^2)}{2E_s} p_{in} \quad (3)$$

where E_s and ν_s are Young's modulus and Poisson's ratio of the target tissue, and p_{in} is the internal pressure of the probe.

The fundamental state equation of contact

$$\delta = \frac{a^2}{2R} + \frac{F(1 - \nu_s^2)}{2E_s a}. \quad (4)$$

C. Stiffness Analysis in Bending and Strain-Stiffening Regime

1) Model I: Bending Stiffness-Dominated Regime: In small indentation ($\delta \sim t$) condition (Mode I), the fluid pressure change is negligible, and the deformation is determined by the membrane's local bending stiffness D and pretension N_0 . Subjected to a distributed load $q(r)$, the governing partial differential equation for the deflection of a thin elastic plate is as follows:

$$D \nabla^4 w(r) - N_0 \nabla^2 w(r) = q(r). \quad (5)$$

The stiffness of K_I in the bending regime is

$$K_I = \frac{\partial F}{\partial \delta} = \frac{4\pi E_m t^3}{3(1 - \nu_m^2) R^2} + 4\pi N_0 \quad (6)$$

where t is the thickness of the membrane, and E_m and ν_m are Young's modulus and Poisson's ratio of the membrane.

In this regime, the interaction force is governed by the bending stiffness of the silicone membrane, and the force-displacement relationship follows a linear response. Therefore, t is designed to be thin and N_0 is low; K_I remains low and constant. This endows the sensor with high conformability, enabling it to conformally contact microtextures.

2) Model II: Strain-Stiffening-Dominated Regime: In large indentation ($\delta \gg t$) condition (Mode II), this deformation process is governed by volume conservation of the internal fluid and fluid-structure interaction of the probe.

As the indentation δ increases, the contact area of the probe

$$A_c(\delta) = \pi a^2 \approx 2\pi R\delta. \quad (7)$$

C_v is the volumetric compliance of the system, and the contact force $F(\delta)$ is derived from the product of the pressure and the contact area

$$F(\delta) = \Delta p(\delta) \cdot A_c(\delta) = \frac{2\pi^2 R^2}{C_v} \delta^3. \quad (8)$$

The stiffness K_{II} in the inflation regime is as follows:

$$K_{II}(\delta) = \frac{\partial F}{\partial \delta} = \frac{6\pi^2 R^2}{C_v} \delta^2. \quad (9)$$

This equation reveals that the stiffness scales quadratically with depth, while the force scales cubically. This capability enables the probe to deform soft surface tissue and detect deep-seated hard nodules.

To verify the stiffness variation of the model, the force-displacement response of the fluid-structure coupled probe during compression against a rigid flat surface is shown in Fig. 2(b). In the initial compression range ($\delta < 0.2$ mm), the response is linear ($R^2 > 0.96$) and the stiffness is 56.36 N/mm, where the force is governed by the bending stiffness of the silicone membrane. As the indentation increases, the force presents a cubic response (Mode II) and a 2.66×10^4 N/mm at 0.88 mm displacement, which has increased by 470× compared with mode I. Moreover, changes in fluid viscosity (1–1000 mPa·s) have minimal impact on stiffness responses since the elastomer membrane dominates the overall stiffness variation (see Fig. 2(c) and Supplementary Material, Fig. S1), while the high specific heat capacity of water prevents heat-induced optical distortion.

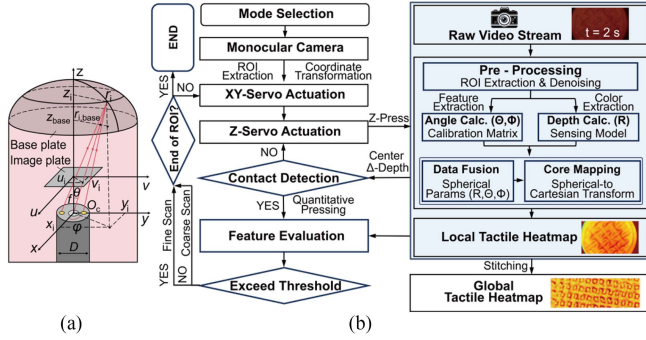


Fig. 3. Schematic of (a) proposed tactile 3-D reconstruction methodology and (b) adaptive robotic scanning framework.

To simulate realistic clinical interactions, 0–10° off-axis indentations (see Supplementary Material, Fig. S2) reveal that increasing inclination reduces sensor stiffness due to asymmetric membrane yielding under lateral shear. This strain-stiffening and incompressibility of the system contribute to a rapid increase in hydrostatic pressure. These results demonstrate the probe’s variable stiffness to passive tuning for different palpation situations.

IV. 3-D SENSING METHODOLOGY AND ADAPTIVE ROBOTIC SCANNING CONTROL FRAMEWORK

After analysis of contact and stiffness tuning mechanisms, we propose a sensing methodology for 3-D reconstruction and an autonomous robotic framework for adaptive scanning. We introduce a geometric mapping algorithm that converts pixel-level chromatic shifts into 3-D topographical data. Moreover, we develop a task-driven, closed-loop scanning strategy that guides the probe to perform continuous, large-area palpation, enabling the accurate analysis of surface texture and localization of subsurface anomalies (see Fig. 3).

A. 3-D Topography Sensing Methodology

To acquire the angular information, we define the optical center of the camera as the origin O_c of the coordinate system. Based on the pinhole camera model, any point P_i on the membrane surface corresponds to a pixel coordinate (u_i, v_i) on the image plane. Given the calibrated focal length f , the directional vector of the incident light ray can be defined in the spherical coordinate system. The angle θ_i with the z -axis and the φ_i are calculated by

$$\theta_i = \arccos\left(\frac{f}{\sqrt{u_i^2 + v_i^2 + f^2}}\right), \quad \varphi_i = \arctan\left(\frac{v_i}{u_i}\right). \quad (10)$$

The distance between the CMOS sensor and the LED is much smaller than the sensing distance r_i ; thus, a coaxial approximation is adopted where the illumination and observation paths are considered collinear. We set $z_{\text{base}} = 5500 \mu\text{m}$ as the base plate for calibration, determined by the camera focal length. Based on our previous sensing model [33], the radial distance r_i is

governed by

$$r_i = k \left(\ln \frac{R_i}{G_i} - \ln \frac{R_{i,\text{base}}}{G_{i,\text{base}}} \right) + \frac{z_{\text{base}}}{\cos \theta_i} \quad (11)$$

where k is the absorption coefficient, and R_i and $R_{i,\text{base}}$ are the red channel intensity of the membrane and base surface. G_i and $G_{i,\text{base}}$ are the green channel intensity of membrane and base surface.

Combining the angular parameters (θ_i, φ_i) and the radial distance r_i , the point cloud (x_i, y_i, z_i) in the Cartesian coordinate system is calculated via spherical transformation

$$\begin{cases} x_i = r_i \sin \theta_i \cos \varphi_i = \frac{r_i u_i}{\sqrt{u_i^2 + v_i^2 + f^2}} \\ y_i = r_i \sin \theta_i \sin \varphi_i = \frac{r_i v_i}{\sqrt{u_i^2 + v_i^2 + f^2}} \\ z_i = r_i \cos \theta_i = \frac{r_i f}{\sqrt{u_i^2 + v_i^2 + f^2}}. \end{cases} \quad (12)$$

This strategy allows measurement of the membrane’s deformation, providing high-fidelity geometric reconstruction.

B. Robotic Scanning Control Framework

To achieve autonomous characterization, we developed a task-driven visuo-tactile control framework. Guided by the contact modeling and target tissue modulus detailed in Section III, the system tunes the z -axis indentation depth to match various diagnostic demands: Mode I (0.3–0.4 mm indentation) targets for surface profiling, and Mode II (1.0–1.2 mm indentation) targets for subcutaneous tumor palpation. For analysis of unknown tissues, a sequential execution of both modes ensures the decoupling of multilayer features.

A monocular camera first identifies the ROI and compensates for the camera-probe spatial offset to align the starting point. During scanning, the z -axis employs closed-loop control based on real-time depth analysis with the task-defined value. Guided by real-time tactile image feedback, the system executes a discrete lift-shift-press trajectory in the xy -plane, employing a step-size adaptive scanning strategy with a 3 mm coarse and 1 mm fine spatial resolution.

During global exploration, the system navigates the ROI following a 3 mm coarse raster scanning trajectory with a scanning rate of 900 mm²/min. Upon reaching the target depth, an adaptive decision-making mechanism evaluates the tactile heatmap. In surface profiling (Mode I), the system calculates the absolute deviations of the raw heatmap to identify surface textures since a smooth surface generates near-zero deviation. In subcutaneous tracking (Mode II), a bilateral filter is first applied to suppress the homogeneous tissue background, after which the absolute deviations are computed to isolate deep-seated local deformations. If this deviation exceeds an empirical threshold (see Supplementary Material, Fig. S3), indicating significant surface textures or deep anomalies, the system autonomously triggers a localized 3×3 fine scanning grid (scanning rate: 120 mm²/min) for high-fidelity reconstruction and targeted boundary delineation. Conversely, if the deviation remains below the threshold, the probe directly proceeds to the next coarse measurement point.

For global reconstruction, we implement a center-crop mosaic approach. The system prioritizes extracting the low-distortion $1.5 \times 1.5 \text{ mm}^2$ central region from the fine tactile images for spatial arrangement, and subsequently inserts the coarse scanning images to fill the remaining gaps. Through the seamless stitching of these multiresolution image segments, a precise global profile reconstruction is achieved.

V. EXPERIMENTAL EVALUATION AND *EX VIVO* VALIDATION

In this section, we conducted a series of experiments to characterize and validate the sensing performance of the sensing probe and robotic palpation system. Experiments aim to demonstrate the measurement fidelity for high-precision 3-D reconstruction, the dual-mode perception capability across different stiffness regimes, and the system integration for automated large-scale scanning tasks.

A. Sensor Characterization

To calibrate the z -axis performance of the sensing probe, vertical indentation experiments were performed by a rigid flat indenter (diameter: 3 mm) installed on a high-precision linear stage, along the z -axis with a step size of $10 \text{ }\mu\text{m}$. A 100×100 pixel window was extracted from the contact region, where the RGB intensities were averaged to generate the representative color values. The logarithmic intensity ratio $\ln(R/G)$ was calculated for each step. Tests across aqueous dye solutions at concentrations of 0.1250%, 0.0833%, and 0.0625% (v/v) indicate that increasing the concentration improves sensitivity, and excessively high concentrations reduce effective measurement ranges. The 0.1250% (v/v) solution was selected as the optimal concentration for subsequent tests. The experimental data reveals a linear ($R^2 > 0.99$) response between the optical data and the z -axis displacement across three repeated trials. This evaluation demonstrates consistent performance of the sensor with an average MAE of $28 \text{ }\mu\text{m}$ [see Fig. 4(a)].

The spatial resolution in the xy -plane is determined by the pixel density of the CMOS sensor. To calibrate this, we employed the linear stage to generate lateral motions, following a procedure similar to the z -axis characterization. The probe was moved laterally from -500 to $500 \text{ }\mu\text{m}$ using the linear stage. By correlating the measured lateral pixel displacements against the known stage coordinates, we established a precise mapping between the image space and physical lengths with an average MAE of $3.1 \text{ }\mu\text{m}$, achieving high-fidelity planar calibration [see Fig. 4(b)]. The reliability of the sensing probe was verified through 500-cycle intensity and force testing, confirming its spatial sensing stability and membrane durability (see Fig. 4(c) and Supplementary Material, Fig. S4). To validate its practical applicability in surgical environments, our tests confirmed that ambient light interference, cleaning procedures, and simulated blood with actual physiological viscosity do not compromise the sensing performance (see Fig. 4(d) and Supplementary Material, Fig. S5).

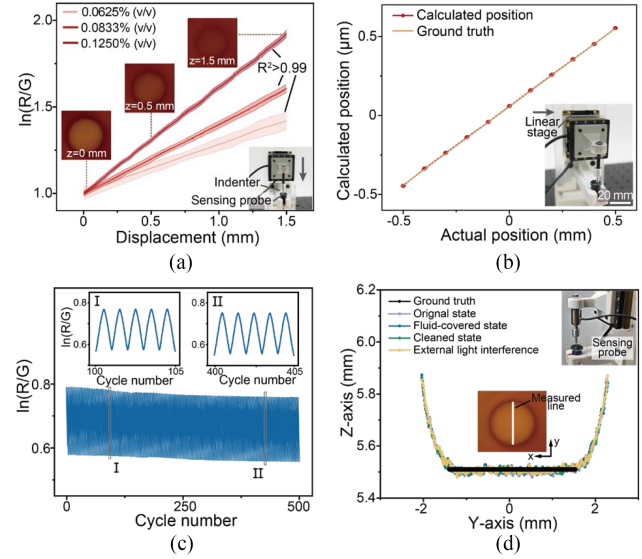


Fig. 4. Characterization of the spatial performance. (a) Z -axis sensing response under different dye concentrations. The central solid line and the shaded region indicate the mean value $\pm$ deviation. (b) Calculated versus actual position of the xy -plane response. (c) Spatial response under cyclic testing. (d) Robust performance under different interferences.

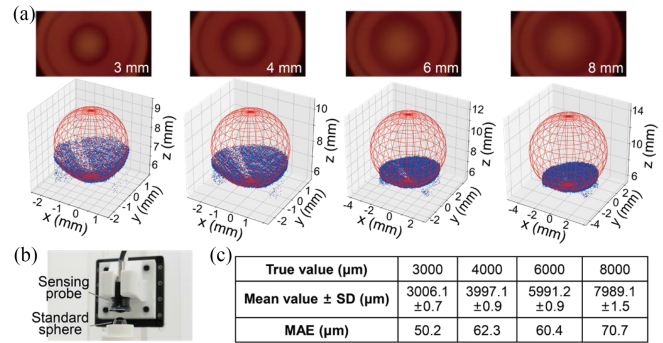


Fig. 5. 3-D reconstruction of standard spheres. (a) Raw images (top) and point clouds (bottom). (b) Test setup. (c) True value, mean value $\pm$ standard deviation, and MAE of reconstruction.

B. Validation of 3-D Reconstruction Accuracy

To evaluate the 3-D reconstruction accuracy, experiments were conducted using standard ceramic spheres with varying diameters (3, 4, 6, 8 mm) as ground-truth targets. The sensing probe was installed on a linear stage to contact the spheres and generate dense 3-D point clouds of the deformed surface [see Fig. 5(a) and (b)]. An algebraic least-squares fitting algorithm was employed to reconstruct the spherical geometry from the point cloud data of the selective region, and each test was repeated three times to generate the average results. The reconstruction error exhibits an increasing trend as the diameter increases. Larger spheres present a flatter contact profile, providing fewer curvature constraints for the algorithm to reconstruct. The deviation remains small (MAE $< 70.7 \text{ }\mu\text{m}$), guaranteeing an accurate sensing performance for tissue analysis [see Fig. 5(c)].

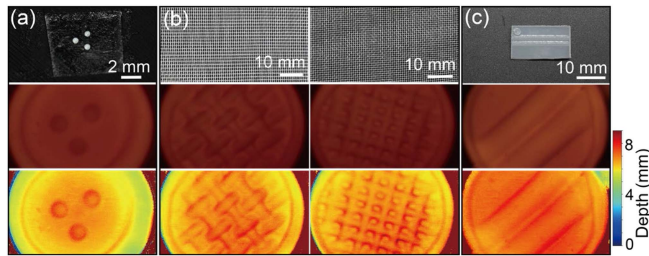


Fig. 6. Surface texture reconstruction (photograph, raw image, and reconstructed profile). (a) Microspheres. (b) Two woven meshes. (c) Silicone semicylindrical ridges.

C. Surface Texture Profiling

Surface texture profiling is important for tissue inspection since many pathological conditions often appear as roughness or granular microstructures on the organ surface. We leveraged the Mode I regime to map the surface textural structures of compliant and rigid targets. The discrete microspheres (diameter: $700\ \mu\text{m}$) were characterized to validate their capacity for analyzing irregular surface protrusions. A 30-mesh (wire diameter: $350\ \mu\text{m}$) and a 50-mesh (wire diameter: $150\ \mu\text{m}$) woven fabric were employed to test the fine-grid spatial resolution for microstructure measurement. Moreover, characterization of a silicone substrate featuring two raised semicylindrical ridges was also performed to resolve the surface morphology with diameters of 1 and 1.1 mm. This ability to capture microscale texture on different moduli and curvatures enables the sensor to detect subtle tissue anomalies on the tissue surface (see Fig. 6).

D. Depth Estimation and Localization of Deep-Seated Tumors

Detecting hard nodules or tumors beneath soft tissue is also critical for clinical palpation. Experiments were performed by embedding different standard spheres (diameter: 3, 4, 6 mm) and depths (1, 2 mm) under a constant indentation (2 mm) into the Ecoflex-0030 silicone bulk. The heatmaps present distinct intensity distributions, where signal area increases with diameter, while peak intensity decreases with depth [see Fig. 7(a)].

Unlike conventional single-point palpation by force sensors, our visuo-tactile sensing unit enables high-density and high-fidelity scanning (see Supplementary Material, Fig. S6). A three-axis Cartesian motion system was employed to achieve a stable, precise scanning process. To validate its capability in characterizing complex simulated tumors (see Supplementary Material, Fig. S7), we embedded stellate and lobulated-shaped inclusions (Dragon Skin 30) into an elastomeric matrix (Ecoflex 30). This tenfold stiffness difference simulates the biomechanics of stiff tumors within healthy tissues. After continuous adaptive scanning and reconstruction image stitching, the resulting tactile heatmap is visualized in Fig. 7(b). The positions and boundaries of the complex tumor phantoms are delineated owing to differences in stiffness. The system achieves a sensitivity exceeding 0.92 and a specificity over 0.99, indicating its reliability in distinguishing tumors from healthy tissues. Furthermore, an

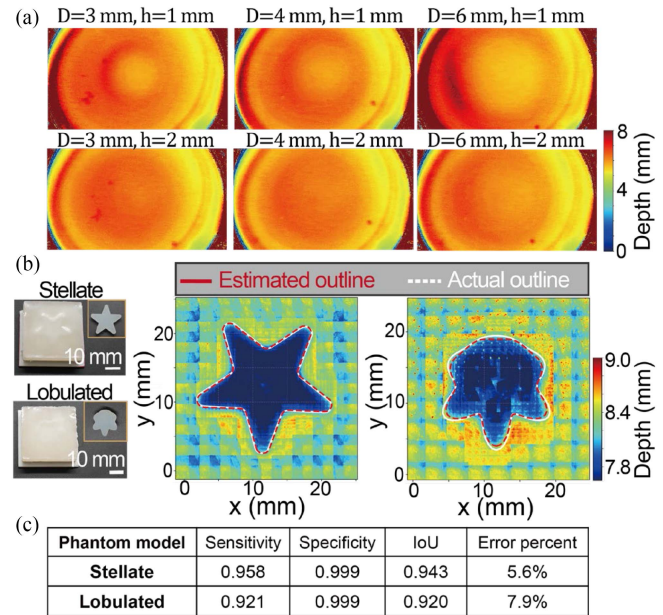


Fig. 7. Characterization of soft complex phantom inclusions. (a) Analysis of spheres with different diameters and embedded depths in soft substrates. (b) Tactile heatmap with actual and estimated outlines of the soft stellate and lobulated inclusions. (c) Evaluation metrics for the phantom tests.

intersection over union (IoU) surpassing 0.92 confirms its high fidelity in precisely delineating tumor boundaries.

E. Integrated Phantom Demonstration

Before evaluating the sensing system in complex biological environments, preliminary validations were conducted using a controlled multilayered silicone phantom (detailed in Supplementary Material, Fig. S8). Building upon these results, to replicate a realistic, dynamic surgical scenario, an *ex vivo* porcine liver tissue block was prepared to simulate multilayered pathologies [see Fig. 8(a)]. A freshly excised porcine liver was procured from a local commercial vendor. A flexible fibrous mesh ($3 \times 25\ \text{mm}$) was adhered to the liver surface to mimic fibrotic stiffening. A planar lobulated silicone structure (diameter: $\sim 7\ \text{mm}$) was fixed 5 mm beneath the surface to represent a complex deep-seated tumor. Furthermore, to emulate adjacent vascular pulsation, a 4-mm silicone tube was embedded 2 cm from the inclusion and actuated with a pulsatile fluid flow at 1 Hz.

The palpation system performed a two-stage scanning: an initial light-touch scan (Mode I) for surface profiling, followed by a deep-perception scan (Mode II) for 3-D tactile reconstruction, to decouple microscale surface textures from the macroscale deep pathology. The fine grid pattern of the mesh is captured on the surface layer via Mode I, presenting its high sensitivity to epidermal irregularities. The 3-D tactile heatmap from Mode II reveals the high-stiffness region and boundary due to the more rigid target. The distinct color in the 3-D heatmap outlines the subcutaneous complex object, demonstrating that the presence of the surface texture and the dynamic perturbations did not

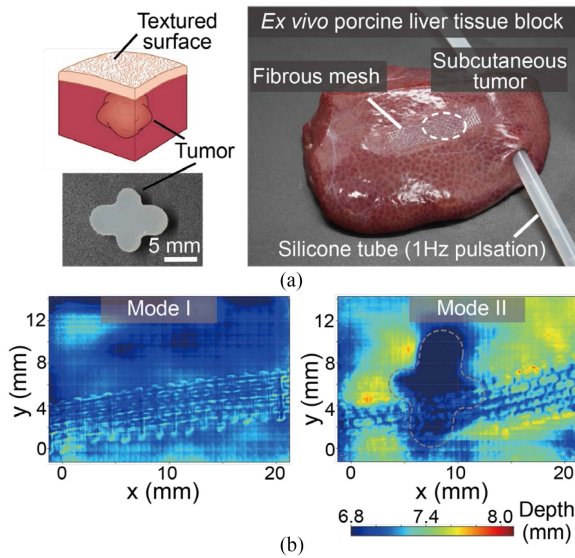


Fig. 8. Reconstruction of the multilayered tissue model. (a) *Ex vivo* porcine liver tissue with surface texture and an embedded tumor model. (b) Tactile heatmaps by two modes.

compromise the analysis of the deep-seated inclusion boundaries [see Fig. 8(b)].

VI. DISCUSSION AND CONCLUSION

In this work, we present a visuo-tactile robotic palpation system for multiscale tissue characterization in minimally invasive environments. The sensing system features a compact visuo-tactile probe (10 mm in diameter, 25 mm in length) that integrates an endoscope and a dyed fluidic medium for 3-D reconstruction, with a fluid-structure coupling mechanism. By exploiting the geometric constraints of an incompressible fluid, the probe achieves adaptive strain-stiffening, transitioning between low-stiffness compliance (Mode I) for surface profiling and high-stiffness indentation (Mode II) for deep tissue penetration.

Combined with 3-D reconstruction and step-size adaptive scanning algorithms, the system achieves an average MAE of 28 μm in the z -axis and 3.1 μm in the xy -plane across three trials. The system profiles microscale surface textures (fine size: 150 μm) with soft and rigid features. It also delineates soft, deep-seated tumors with complex stellate and lobulated morphologies, achieving high sensitivity (> 0.92) and IoU (> 0.92). Moreover, the system's clinical potential was validated on an *ex vivo* porcine liver model. By integrating light touch and deep indentation, the platform decoupled multilayered pathologies, characterizing superficial fibrosis and underlying inclusions.

While the proposed robotic palpation system demonstrates robust performance, there remain certain limitations to be addressed. Future research will investigate the system's quantitative detection limits across lesions with varying stiffness contrasts and ambiguous margins. Furthermore, comprehensive system-level evaluations will be conducted to assess long-term repeatability and the inherent tradeoffs between scanning speed

and reconstruction accuracy. Finally, future work will also integrate machine learning to automate pathology characterization and incorporate real-time motion compensation for adapting to dynamic *in vivo* environments, establishing this robotic palpation system as a robust quantitative tool for intraoperative diagnosis.

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